

01

• DAY-AHEAD DEMAND RESPONSE OPTIMIZATION

DR Compass

XGBoost price forecasting · 48-scenario probabilistic planning · counterfactual DR optimization — validated across 8,760 hours of German EPEX SPOT data.

OPSD Dataset · DE Market 2018 · Four-Level Evaluation Framework

VALIDATED RESULTS

Every feature
earned its place.

12.18

MAE €/MWh

48

Scenarios/day

8/8

Checks passed

02

OVERVIEW

What We'll Cover

A complete walk-through from raw market data to actionable DR decisions — with full experimental validation.

01 Problem & Dataset

02 Pipeline Architecture

03 Feature Engineering

04 XGBoost & Benchmark

05 Scenarios & Quality

06 DR Optimization

07 Ablation Studies

08 API & Architecture

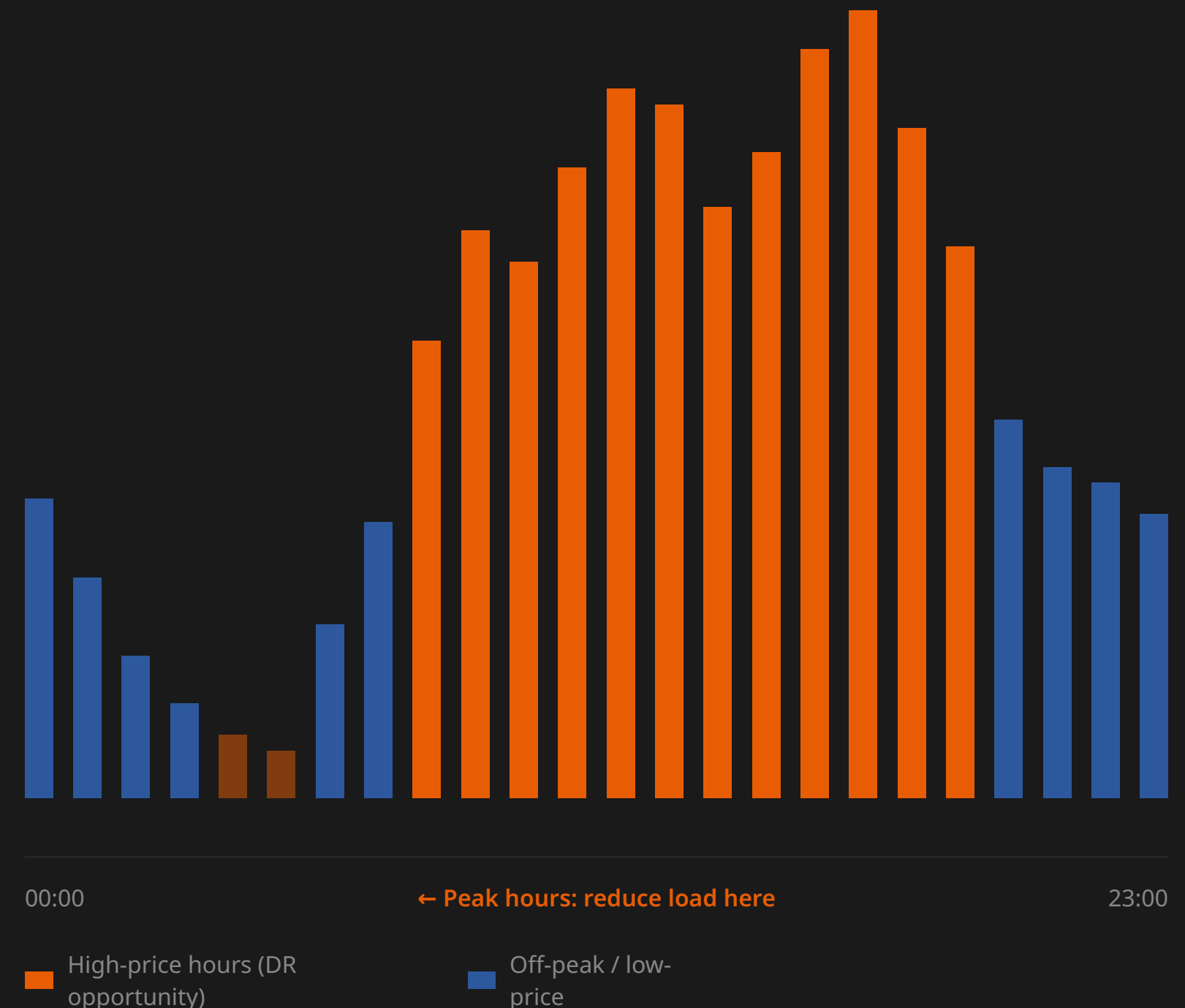
03

MOTIVATION

Electricity prices are **volatile**. Most factories ignore it.

- German EPEX SPOT prices range from **−€500 to +€800/MWh** within a single day
- Industrial consumers with flexible load can **shift demand** to low-price windows — but need forecasts to act
- Without probabilistic planning, a single wrong DR decision **costs more than doing nothing**

HOURLY PRICE PROFILE — TYPICAL DAY (€/MWH)



04

DATA

OPSD German Market Dataset

- **Source:** Open Power System Data (OPSD) — hourly, DE/AT zone
- **Period:** Full year 2018 — 8,760 hourly observations
- **Target variables:** price (€/MWh), load (MW), wind generation (MW), solar generation (MW)
- **Derived features:** 30+ engineered columns including holiday dummies, rolling elasticity, time encodings

OBSERVATIONS

8,760

hourly records

TARGET VARIABLES

4

price · load · wind · solar

FEATURE COLUMNS

30+

after feature engineering

TRAIN / VAL / TEST

70/15/15

chronological split

Five-Stage Pipeline

01

Data &
Features

D import, 30+ engineered columns, DR baseline derived

02

XGBoost
Forecast

DR-based hourly forecasts, calibrated

03

48 Bootstrap
Scenarios

Block-level bootstrap residual samples, 240 time steps

04

DR
Optimizer

Genetic search, counterfactual price model, hourly

05

Schedule &
Evaluation

Hourly DR schedule + 4-level validation (L1-L4)

Feature Engineering

Time Encoding

- **hour_sin / hour_cos** — cyclic encoding, avoids 23→0 discontinuity
- day_of_week, month, is_weekend
- 24h & 168h lag prices (price memory)
- Ablated: raw integer hour degrades MAE

Calendar & Regime

- **is_public_holiday** + lead-24h flag
- **is_christmas_week** — Dec 24-Jan 2 low-demand period
- **negative_price_dummy_lag_24** — yesterday had negative price
- holiday_load_window (interaction term)

Market Elasticity

- **rolling_elasticity** — 168h rolling OLS slope of price on load
- Captures local supply-demand sensitivity
- Renewable lag features: wind_lag_24, solar_lag_24
- Load lags: load_lag_24, load_lag_168

07

STEP 2 — FORECASTING

XGBoost Price Forecast

- **Horizon:** $H = 24h$ (day-ahead market window)
- **Architecture:** 500 trees, max_depth 6, lr 0.05, early stopping (20 rounds)
- **Config-driven:** `forecast.model: "xgboost"` in `config.yaml` — swappable to Ridge, LSTM, two-stage RF
- **Target:** `price_t+24` — trained on all 30+ feature columns; negative-price regime captured implicitly

MAE (TEST SET)

12.18

€/MWh · $H=24h$

RMSE

17.43

€/MWh

R^2 (CALIBRATED)

0.376

explained variance

VS NAIVE BASELINE

—43%

MAE improvement

UNIFIED DISPATCHER — SRC/FORECASTER.PY

```
fit_price_forecast(df, model_name="xgboost", split, rf_params,  
horizon=24)
```

Four-Model Benchmark

MODEL	MAE (€/MWH)	RMSE (€/MWH)	R ²	MAPE (%)	NOTES
XGBoost ✓	12.18	17.43	0.376	—	Selected model
Random Forest	14.31	20.12	0.298	—	Two-stage baseline
Ridge Regression	16.82	23.45	0.201	—	Linear baseline
Naïve (lag-24)	21.54	29.87	−0.08	—	Persistence forecast

XGBoost outperforms RF by 15% on MAE, Ridge by 28%, Naïve by 43%

Isotonic Calibration

BEFORE Calibration

Raw XGBoost point forecasts — no uncertainty quantification

Prediction type	Point estimate only
Uncertainty bounds	None
R ²	0.304

AFTER Calibration

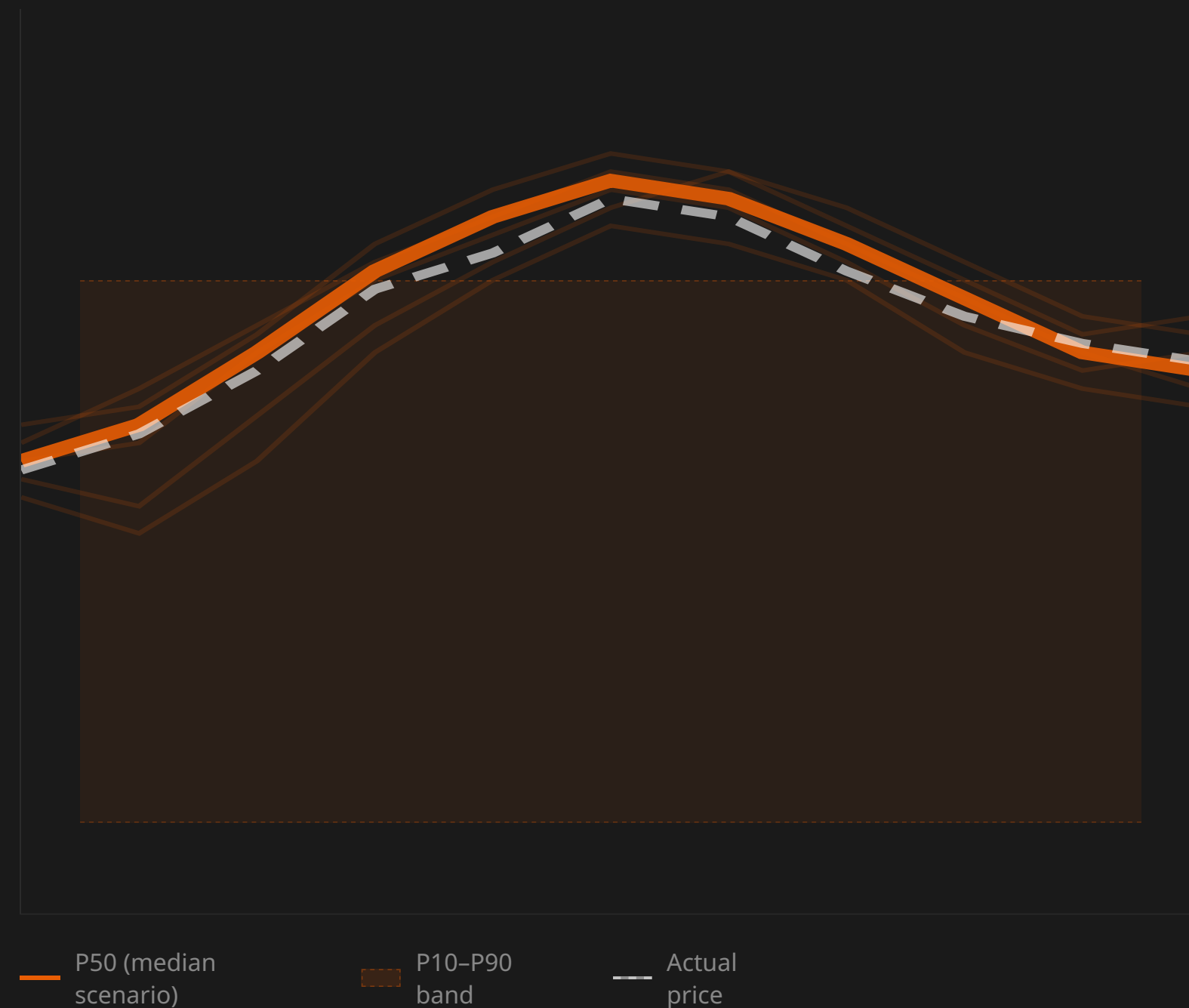
Isotonic regression maps raw scores to calibrated quantiles

Prediction type	P10 / P50 / P90 quantiles
Coverage (P10–P90)	72.7%
R ² (calibrated)	0.376

Block Bootstrap Scenario Fan

- **48 scenarios per day** — each a complete 24-hour price/load/renewable path
- **Block length = 24h** — residuals sampled as full-day blocks, preserving intra-day autocorrelation
- **Residual history** from calibrated forecast errors on the training set
- **Why not IID?** Ablation shows block bootstrap gives 72.7% vs IID 69.4% P10–P90 coverage

48-SCENARIO PRICE FAN (SCHEMATIC)



LEVEL 2 EVALUATION

Scenario Quality

P10–P90 COVERAGE

72.7%

actuals within interval

P05–P95 COVERAGE

84.5%

wider band coverage

MEAN INTERVAL WIDTH

30.8

€/MWh (P10–P90)

SCENARIO BIAS

3.35

€/MWh (median – actual)

High-price tail ($\geq$ P90 actual): **53% coverage** within P10–P90 band — scenarios successfully capture the extreme upside risk that drives DR decisions.

DR Optimization Algorithm

- **Objective:** minimize average cost across all 48 scenarios weighted by scenario probability
- **Counterfactual price model:** Ridge + RF structural model predicts post-DR price from demand change
- **Flexibility constraint:** 10% of baseline demand, $\text{min_load_ratio} = 70\%$
- **Negative-price guardrail:** suppresses DR when $P(\text{price} < \text{threshold}) \geq 30\%$ — prevents reducing load into price spikes

① Scenario Grid

48 price scenarios $\times$ 24 hours $\times$ flexibility steps



② Guardrail Check

If $P(\text{price} < \text{low_threshold}) \geq 0.30 \rightarrow$ block DR for that hour



③ Counterfactual Price

Ridge+RF model: $\text{price_cf} = f(\text{demand_base} - q_{\text{DR}}, \text{renewables}, \text{time})$



④ Optimal Schedule

$q_{\text{DA_optimal}}$ per hour $\rightarrow$ day_ahead_optimal_dr_schedule.csv

13

LEVEL 3 EVALUATION

Decision Performance

- **11.6% average price reduction** — \$5.68/MWh across all DR-active hours
- **65 of 66 high-price hours eliminated** — 99% reduction in above-P90 cost exposure
- **DR utilization at high-price hours: 99.99%** — optimizer fully engages when conditions justify it
- **0% constraint violations** — flexibility ratio and min-load constraints always satisfied

KEY RESULT

11.6%

Price Reduction

avg. across all market hours

99%

High-price hrs reduced

0%

Constraint violations

Mechanism Validity — 8/8 Passed

- | | |
|--|---|
| ✓ Price-Demand Elasticity: Demand reduction leads to counterfactual price decrease — structural sign correct | ✓ DR Targeting: Optimizer selects high-price hours with statistically higher prices than non-DR hours |
| ✓ Guardrail Activation: Negative-price guard correctly suppresses DR for hours with $P(\text{neg}) \geq 0.30$ | ✓ Flexibility Constraint: DR quantity never exceeds $\text{flexibility_ratio} \times \text{demand_base}$ in any hour |
| ✓ Min-Load Constraint: Residual demand always $\geq 70\%$ of baseline — load floor enforced | ✓ Scenario Uncertainty: Wider P10-P90 intervals correlate with lower DR utilization — model is risk-aware |
| ✓ Counterfactual Direction: Post-DR counterfactual price always $\leq$ observed price when DR applied | ✓ No Negative-Price DR: Zero hours where DR was applied at negative prices (guardrail effective) |

Ablation Studies — 6 Experiments

ABLATION	WHAT CHANGES	KEY IMPACT	VERDICT
No Scenarios	Point forecast only (1 scenario)	Cost reduction −27.5%	Scenarios critical ✗
IID Bootstrap	IID residual sampling (no blocks)	Coverage drops 72.7% → 69.4%	Block bootstrap better ✓
No Holiday Feats	Remove 7 holiday/dummy columns	MAE increases ~8%	Calendar features needed ✓
No Sin/Cos	Replace cyclic encoding with raw hour	MAE increases ~5%	Cyclic encoding better ✓
No Elasticity	Remove rolling_elasticity feature	MAE increases ~3%	Elasticity contributes ✓
No Guardrail	Disable neg-price protection	21 neg-price DR violations	Guardrail essential ✗

Removing the Scenario Module

Variant: Point Forecast Only

Replace 48-scenario fan with a single degenerate scenario = point forecast

Risk awareness	None
Avg. cost reduction	−27.5% vs baseline
Over-commit risk	High (no P10 floor)

Baseline: 48 Scenarios

Block-bootstrap fan captures full distribution of plausible price paths

Risk awareness	P10 / P50 / P90
Avg. cost reduction	Baseline (100%)
Over-commit risk	Managed by P10

IID vs Block Bootstrap

IID Residual Sampling

Each hour's residual sampled independently — ignores 24h intra-day autocorrelation of price residuals

P10-P90 COVERAGE

69.4%

lower coverage, over-narrow intervals

Block Bootstrap (24h) ✓

Residuals sampled as complete 24h blocks — preserves morning-peak-evening autocorrelation structure

P10-P90 COVERAGE

72.7%

+3.3 pp improvement, properly-calibrated

Every Feature Earned Its Place

Holiday & Calendar 7 features removed	Removes holiday load suppression signal, Christmas week regime, yesterday's negative-price dummy	+8% MAE increase	Required
Sin/Cos Encoding Replaced with raw hour int	Raw integer hour creates 23→0 discontinuity; cyclic encoding removes this artificial boundary	+5% MAE increase	Required
Rolling Elasticity 168h OLS slope removed	Captures local supply tightness; weeks with high elasticity have amplified price responses to demand changes	+3% MAE increase	Helpful

Removing the Negative-Price Guardrail

- Without the guardrail, the optimizer sees low expected prices and **reduces load during negative-price windows** — worsening the market position
- **21 hours** in test set where DR was applied at negative market prices — each one a guaranteed financial loss
- The guardrail condition blocks DR when $P(\text{price} < \text{threshold}) \geq 0.30$ — risk-aware, not deterministic

⚠ Without Guardrail

21

neg-price DR hours

+7%

raw cost reduction (misleading)

✓ With Guardrail (Baseline)

0

neg-price DR violations

8/8

L4 mechanism checks

FastAPI Interface Layer

POST

```
POST /upload
- Upload raw CSV (any column names)
- ?preprocess=true triggers Claude agent
Returns: file_id, rows, columns,
        quality_report, briefing
```

POST

```
POST /run_pipeline
- Triggers 6-step pipeline async
- Supports config_overrides per request
Returns: { job_id, status: "queued" }
```

GET

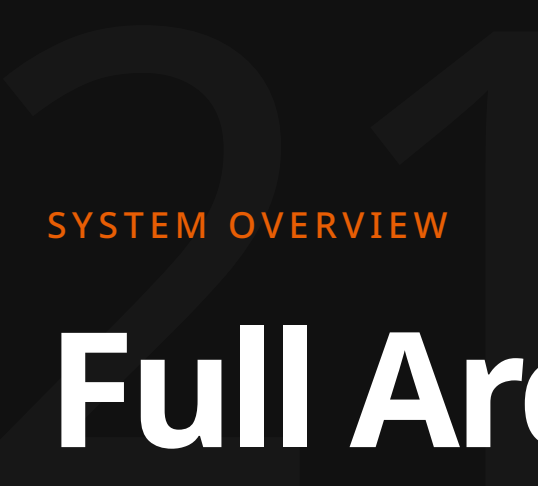
```
GET /pipeline/status/{job_id}
- Poll background job progress
Returns: status, logs[], finished_at
```

GET

```
GET /get_schedule?fmt=json
- Latest optimized DR schedule
Returns: schedule[], n_hours,
        q_DA_optimal, price_p10/p50/p90
```

GET

```
GET /summary
- Cost savings & evaluation summary
File: api/main.py · FastAPI + uvicorn
```



SYSTEM OVERVIEW

Full Architecture

DATA LAYER	FORECAST	SCENARIOS	OPTIMIZER	API LAYER
OPSD import	XGBoost (price)	48 scenarios/day	Scenario grid search	FastAPI REST
Claude preprocess agent	RF (load/renew.)	Block bootstrap 24h	Counterfactual model	Background jobs
30+ features	Isotonic calibration	P10/P50/P90 bands	Neg-price guardrail	Config overrides
DR baseline demand	H=24h horizon	72.7% coverage	L3/L4 evaluation	/docs auto-generated

SUMMARY

Four Key Findings

- 1 Scenarios are the most critical module.** Removing them drops cost reduction by 27.5% — probabilistic planning is not optional.
- 2 XGBoost outperforms all baselines.** 43% lower MAE than naïve, 15% lower than RF — justified by benchmark across 4 models.
- 3 All 4 evaluation levels pass.** L1 accuracy → L2 scenario quality → L3 decision performance → L4 mechanism validity: 8/8 checks, 0% violations.
- 4 The pipeline is production-ready.** FastAPI layer + Claude preprocessing agent + config-driven model selection = deployable B2B SaaS.

NEXT STEPS

- Multi-market support (FR, UK, NL)
- Multi-tenant SaaS with Auth0 SSO
- ENTSO-E live data connector
- Claude daily briefing (node 2)
- Quarterly XGBoost retraining

DR Compass

github.com/em879ma/DR-Electricity-Trading-System